



**NEXORD**

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# COMPUTATIONAL ORDINAL SOCIOMETRY AND SOCIO-THERMODYNAMIC DIAGNOSTICS (NEXORD PLATFORM)

## Scientific Manifesto, Theoretical-Methodological Framework, and Institutional Governance Guidelines

«Hacia una Física de lo Grupal\_JMC»

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**Official Web Platform & Interactive Ecosystem:** <https://josemancor.github.io/nex-ord-web/>

**Official Repository & Source Code:** <https://github.com/josemancor/nex-ord-web>

**Master Brand / Ecosystem:** VISORD 3D/4D (Visual Ordinal Sociometry & Socio-Thermodynamic Diagnostics)

**Platform Supported Languages:** Spanish (ES), English (EN), French (FR), German (DE), Italian (IT), Portuguese (PT), Russian (RU), Chinese (ZH).

## Executive Abstract

Computational Ordinal Sociometry (NEXORD) is, in essence, an updated extension of the foundational insights of Jacob Levy Moreno, creator of Sociometry, and Kurt Lewin, pioneer of field theory—two seminal figures in Social Psychology. Historically, group dynamics research has been constrained by the methodological and technical limitations of its founding era. This Manifesto is grounded in:

**1)** The conception of the ‘relational tie’, born of human interaction, as the core constituting the existential reality of groups. ‘Giving and Receiving’ are indissociable force pairs that sustain the necessary dialectical exchange of affect, ideas, intentions, resources, and collective actions toward shared goals. Personal and group health, wellbeing, success, and viability fundamentally depend on the continuous homeostatic balance established across interpersonal exchanges of acceptance and conflict.

**2)** The refined quantification of preferential relational order through the inverse rank function, together with contemporary theoretical, methodological, and technological developments, defines the genesis of NEXORD as an overarching sociometric platform operationalizing Moreno’s and Lewin’s core intuitions.

**3)** The continuous metric emerging from this quantification is formally articulated, through matrix and tensorial algebra, within the Moreno-rooted relational tetragram ( $E, R, pE, pR$ ) across  $3^4 = 81$  dyadic configurations operationalized via the SMib/SMIa matrix duality, translating subjective micro-interaction into computable topology.

**4)** The articulation of the 4 elemental densities and their analytical extension to 18 derived relational densities (18D space) from the Q81 lattice, quantifying the mass and flux of exchanges. This continuous gradient models the collective as a socio-thermodynamic system governed by equilibrium regimes and relational friction ( $\Phi$ ), where dialectical friction acts as a necessary condition for systemic health and adaptability against groupthink entropy.

**5)** Simultaneous projection across dual factorial spaces (CA  $\parallel$  PCA), wherein actors emerge not as static scalar points, but as dyadic kinematic dipoles ( $DA \leftrightarrow RECEIVE$ ) whose centroid analytically equilibrates emission and reception. This formulation reveals the gravitational dominance of the primary relational axis ( $PC_1 \gg PC_2$ ) and enables scale-invariant geometric comparison of group topologies across Grassmann manifolds  $\operatorname{Gr}(k, N)$ .

**6)** The versatility of the multidimensional design ( $G \times T \times C$ ), enabling integrated transversal, longitudinal, and criterial comparisons (intra- and inter-group). This architecture is enhanced by the interactive VISORD visualization modules, facilitating the monitoring of sociometric structures and processes emerging from the

analyses.

7) NEXORD's effective ethical commitment to a strictly constructive and non-punitive sociometric praxis, ensuring absolute privacy protection by design (*Privacy by Design*). This framework mandates institutional oversight and the explicit deontological commitment of the accredited Principal Investigator (PI) as sole interlocutor, under Open Science standards.

**Keywords:** NEXORD Platform, VISORD, Computational Ordinal Sociometry, Preventive Sociometry, Breve Format, Inverse Rank Function, Relational Densities (18D), Metaperception, Dual Factorial Planes, Grassmann Manifolds, Groupthink, Ostracism, Ethics by Design, Open Science.

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## **| 1. Epistemological Foundations**

### **1.1. Overcoming the Classical Limit & the Ontology of the Relational Tie**

NEXORD's objective is the methodological updating of social psychology traditions: translating into computational formalisms the foundational insights that neither Jacob L. Moreno nor Kurt Lewin could operationalize with the technical tools of their era. Moreno introduced the concept of the social atom, but manual computation restricted sociometry to fixed binary quotas (typically three choices and three rejections). Lewin formulated psychological field theory and social topology, but lacked continuous mathematical operators to quantify relational distances without distorting the psychological nature of interpersonal ties. NEXORD places active relational interaction at the ontological core of group reality: the collective is not an aggregate of isolated nodes, but a dynamic field of forces in continuous emergence.

### **1.2. Inverse Rank Quantification**

The mathematical foundation enabling the transition to continuous metrics is the quantification of hierarchical order through the inverse rank function:

#### Formula 1: Relational Decay

where  $r_{ij}$  in  $\{1, 2, \dots, N\}$  represents the strict ordinal rank assigned by sender  $i$  to receiver  $j$ , and  $\gamma > 0$  governs the field's openness and relational decay rate. This formulation translates qualitative ordering into continuous relational values, allowing dyadic ties to be operated as measurable forces projectable into multidimensional spaces. This formal principle aligns with statistical frameworks for rank-order relational data (Krivitsky & Butts, 2017) and psychometric findings on measurement degradation in truncated discrete nominations (Terry, 2000).

### 1.3. The Fundamental Axiom: GIVING and RECEIVING as an Indissociable Circuit

Emitting choices (GIVING) and receiving collective evaluations (RECEIVING) do not represent independent variables or isolated traits: they are immanent and indissociable moments of a single continuous relational circuit. The functional health or strain of a group does not reside within isolated members, but in the fluidity, symmetry, and friction of their mutual exchanges. An actor may emit positive choices while facing systematic collective exclusion; detaching these two polar moments produces incomplete diagnostics blind to latent

structural isolation.

## 2. Capture Protocol: BREVE Format, Matrix Duality & Diagonal Closure

### 2.1. Forced Drag without Ties & Matrix Duality: A(i, rank) vs. B(i, j)

In the BREVE Format, participants view a randomized peer list to eliminate order biases. Through a forced drag-and-drop task, they rank all group members between maximum preference and rejection poles. The zero-ties rule ( $r_{ij} \neq r_{ik}$ ) eliminates acquiescence bias and ensures complete ranking. This capture is mathematically formalized into two complementary matrices:

- Matrix A(i, rank) BREVE:** Represents direct ordinal sequences emitted by each participant. Rows represent senders  $i$ , columns represent ordinal ranks  $k \in \{1st, ..., Nth\}$ , and cell contents record the peer placed at that rank:  $A(i, k) = j$ .
- Matrix B(i, j) RANK DISTRIBUTION:** Square sociomatrix ( $N \times N$ ) where rows represent senders  $i$ , columns represent receivers  $j$ , and cell values record the assigned numerical rank:  $B(i, j) = k \Leftrightarrow A(i, k) = j$ .

Table 1A: Factual Formalization of Group A ( $G_1$ ,  $N = 6$ , Bipolar  $3+3$ ) — BREVE Format, Classical Moreno, and Metaperceptions

| Node ID | 1st | 2nd | 3rd | 4th | 5th | 6th | Choices (E) | Rejections (R) | Expected Choices (pE) | Expected Rejections (pR) | $\alpha$ E\$ (Closest) | $\omega$ E\$ (Distant) |
|---------|-----|-----|-----|-----|-----|-----|-------------|----------------|-----------------------|--------------------------|------------------------|------------------------|
| 1A      | 2A  | 1A  | 3A  | 5A  | 4A  | 6A  | 2A, 3A      | 4A, 6A         | 2A, 3A                | 4A, 6A                   | 2A                     | 2A $\rightarrow$ 1A    |
| 2A      | 1A  | 2A  | 3A  | 4A  | 6A  | 5A  | 1A, 3A      | 5A, 6A         | 1A, 3A                | 5A, 6A                   | 1A                     | 1A $\rightarrow$ 2A    |
| 3A      | 1A  | 2A  | 3A  | 6A  | 5A  | 4A  | 1A, 2A      | 4A, 5A         | 1A, 2A                | 4A, 5A                   | 1A                     | 1A $\rightarrow$       |

|    |    |    |    |    |    |    |        |        |        |        |    |    |          |
|----|----|----|----|----|----|----|--------|--------|--------|--------|----|----|----------|
|    |    |    |    |    |    |    |        |        |        |        |    |    | 2A       |
|    |    |    |    |    |    |    |        |        |        |        |    |    | 5A \$→\$ |
| 4A | 5A | 4A | 6A | 2A | 1A | 3A | 5A, 6A | 1A, 3A | 5A, 6A | 1A, 3A | 5A | 4A |          |
|    |    |    |    |    |    |    |        |        |        |        |    |    | 4A \$→\$ |
| 5A | 4A | 5A | 6A | 1A | 3A | 2A | 4A, 6A | 2A, 3A | 4A, 6A | 2A, 3A | 4A | 5A |          |
|    |    |    |    |    |    |    |        |        |        |        |    |    | 4A \$→\$ |
| 6A | 4A | 5A | 6A | 3A | 2A | 1A | 4A, 5A | 1A, 2A | 4A, 5A | 1A, 2A | 4A | 5A |          |

Table 1B: Instrumental SMIb Sociomatrix of Group A (\$G\_1\$, \$N = 6\$) — Dyadic Tetragram \$(A\_1 A\_2 A\_3 A\_4 = pDA \cdot DA \cdot RECIBE \cdot pRECIBE)\$

| \$G_1\$ | 1A   | 2A   | 3A   | 4A   | 5A   | 6A   |
|---------|------|------|------|------|------|------|
| 1A      | 0AA0 | ABBA | ABBA | 0ab0 | 0ab0 | 0ab0 |
| 2A      | BAAB | 0AA0 | ABBA | 0ab0 | 0ab0 | 0ab0 |
| 3A      | BAAB | BAAB | 0AA0 | 0ab0 | 0ab0 | 0ab0 |
| 4A      | 0ba0 | 0ba0 | 0ba0 | 0AA0 | ABBA | ABBA |
| 5A      | 0ba0 | 0ba0 | 0ba0 | BAAB | 0AA0 | ABBA |
| 6A      | 0ba0 | 0ba0 | 0ba0 | BAAB | BAAB | 0AA0 |

**Table 1C: Integrated SMIa Sociomatrix of Group A (\$G\_1\$, \$N = 6\$) — Canonical Q81 Lattice**

| \$G_1\$ | 1A                     | 2A                     | 3A                     | 4A     | 5A     | 6A     | Subgroup Diagnostic                                                           |
|---------|------------------------|------------------------|------------------------|--------|--------|--------|-------------------------------------------------------------------------------|
| 1A      | $i\mathcal{C}?! \quad$ |                        | $[Rr]$                 | $[Rr]$ | $[Rr]$ |        | Cohesion $\mathcal{S}\{1A, 2A, 3A\}$ / Opposition $\mathcal{S}\{4A, 5A, 6A\}$ |
| 2A      |                        | $i\mathcal{C}?! \quad$ |                        | $[Rr]$ | $[Rr]$ | $[Rr]$ | Cohesion $\mathcal{S}\{1A, 2A, 3A\}$ / Opposition $\mathcal{S}\{4A, 5A, 6A\}$ |
| 3A      |                        |                        | $i\mathcal{C}?! \quad$ | $[Rr]$ | $[Rr]$ | $[Rr]$ | Cohesion $\mathcal{S}\{1A, 2A, 3A\}$ / Opposition $\mathcal{S}\{4A, 5A, 6A\}$ |

|    |      |      |      |      |                                                         |
|----|------|------|------|------|---------------------------------------------------------|
| 4A | [Rr] | [Rr] | [Rr] | i¿?! | Cohesion $\{4A, 5A, 6A\}$ / Opposition $\{1A, 2A, 3A\}$ |
| 5A | [Rr] | [Rr] | [Rr] | i¿?! | Cohesion $\{4A, 5A, 6A\}$ / Opposition $\{1A, 2A, 3A\}$ |
| 6A | [Rr] | [Rr] | [Rr] | i¿?! | Cohesion $\{4A, 5A, 6A\}$ / Opposition $\{1A, 2A, 3A\}$ |

Table 1D: Factual Formalization of Group B (\$G\_2\$, \$N = 6\$, Systematic Ostracism of 6B) — BREVE Format and Inbound Status (\$SRC\$)

| ID | 1st | 2nd | 3rd | 4th | 5th | 6th<br>(Rejection) | Node 6B in<br>SMIa                                                       | Inbound Status<br>(\$SRC\$) | Structural Diagnosis                         |
|----|-----|-----|-----|-----|-----|--------------------|--------------------------------------------------------------------------|-----------------------------|----------------------------------------------|
| 1B | 2B  | 1B  | 3B  | 4B  | 5B  | 6B                 | $\$1B \rightarrow 6B\$$ :<br>[Rr]                                        | $\$+0.875\$$                | Cohesive Attractor Leader                    |
| 2B | 1B  | 2B  | 3B  | 5B  | 4B  | 6B                 | $\$2B \rightarrow 6B\$$ :<br>[Rr]                                        | $\$+1.250\$$                | Primary Attractor Leader                     |
| 3B | 1B  | 2B  | 3B  | 4B  | 5B  | 6B                 | $\$3B \rightarrow 6B\$$ :<br>[Rr]                                        | $\$+0.625\$$                | Cohesive Core                                |
| 4B | 3B  | 4B  | 1B  | 2B  | 5B  | 6B                 | $\$4B \rightarrow 6B\$$ :<br>[Rr]                                        | $\$+0.125\$$                | Integrated Neutral                           |
| 5B | 2B  | 3B  | 5B  | 1B  | 4B  | 6B                 | $\$5B \rightarrow 6B\$$ :<br>[Rr]                                        | $\$+0.125\$$                | Integrated Neutral                           |
| 6B | 1B  | 2B  | 6B  | 3B  | 4B  | 5B                 | $\$6B \rightarrow$<br>$\backslash\text{text{Group}}\backslash$<br>$\$$ : | $\$-1.500\$$                | Unanimous Ostracism<br>(Relational Asphyxia) |

| ID | BREVE (I) |    |  |
|----|-----------|----|--|
|    | 1º        | 2º |  |
| 1A | 2A        | 1A |  |
| 2A | 1A        | 2A |  |

*Figure 1: BREVE Input Architecture and Derived SM Ib / SM Ia Matrices (Canonical  $N = 6$  Scenario, Bipolar Group  $G_1$ ). Left Panel: Forced ranking Matrix  $A(i, \text{rank})$  1st to 6th, bijective Moreno transcription ( $E, R, pE, pR$ ), and triangulated metaperceptions ( $\alpha E, \omega E$ ). Center Panel: Instrumental SM Ib sociomatrix of dyadic tetragrams ( $A_1 \cdot A_2 \cdot A_3 \cdot A_4$ ). Right Panel: Integrated SM Ia sociomatrix in  $Q_{81}$  lattice.*

## 2.2. Reflexive Diagonal Closure & Core Sensors

NEXORD resolves the classical sociometric diagonal omission (traditionally empty,  $r_{ii} = \text{empty}$ ) by requiring participants to self-anchor within the ranking ( $r_{ii}$  in  $\{1, \dots, N\}$ ). This reflexive closure equips the system with three analytical sensors:

- **IDA (Self-Rank Misalignment Index):** Angular difference between self-image ( $r_{ii}$ ) and received peer consensus ( $R_{\text{ext}_i}$ ):

Formula 2: Self-Image Discrepancy IDA

where  $IDA < 0$  marks status overestimation,  $IDA > 0$  indicates self-deprecation or timidity, and  $IDA = 0$

reflects balanced social calibration. At the collective level, group deviation is computed quadratically:

Formula 3: Group Discrepancy IDA\_group

- **ITA (Self-Rank Tension Index):** Evaluates structural strain arising when multiple members claim rank one ( $r_{ii} = 1$ ).
- **Cognitive Partitioning  $P(k)$ :** By positioning oneself at rank  $k$ , the combinatorial space partitions into  $P(k) = (k-1)!(N-k)!$ , formalizing that interpersonal evaluations are projected upward or downward from an initial self-anchor:

Formula 5: Combinatorial Anchoring  $P(k)$

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## 3. Relational Dynamics: Field Regimes, Friction & Viability

### 3.1. Relational Friction Coefficient ( $\Phi$ ) and Structural Cohesion (ICE)

Collective internal resistance and misalignment are synthesized in the Relational Friction Coefficient ( $\Phi$ ) and its structural complement, the Structural Cohesion Index (ICE):

$$\Phi = \sqrt{0,5 \cdot IT}$$

Formula 4: Relational Friction and Structural Cohesion ICE

where moderate values of  $\Phi$  reflect adaptive vitality, while extreme values indicate structural strain or polarization risks.

### 3.2. Psychologization of Field Regimes & Groupthink Prevention

Group dynamics should not be framed as inanimate physics: collective states reflect psychosocial regimes of interpersonal articulation:

- **Regime of Structural Rigidity and Inertia:** Characterized by frozen hierarchies, static roles, and reduced communication permeability.
- **Regime of Fluidity and Adaptive Cohesion:** Characterized by open relational homeostasis, where preferences and constructive feedback circulate flexibly.
- **Regime of Dispersion and Anomie:** Characterized by low coupling density, disengagement, and weak collective grounding.
- **Regime of Effervescence and Collective Fusion:** Characterized by shared emotional activation

and intense interactive mobilization.

Within this framework, NEXORD questions the assumption that optimal functioning equals absolute cohesion ( $ICE = 1.0$ ,  $\Phi = 0$ ). A collective devoid of friction represents a vulnerable system prone to Groupthink (Janis, 1972), where consensus pressures suppress divergent perspectives. Moderate friction and transactional gradients supply the necessary impetus for organizational learning and adaptive plasticity.

3.3. The Living Puzzle: Unjamming vs. Dignified Ecosystemic Relocation

Unlike models addressing friction through punitive exclusion or individual blaming, NEXORD views the collective as a living puzzle. An actor experiencing relational tension is not inherently deficient: they often encounter structural role misalignments or flow asymmetries within their local network. Diagnostic intervention prioritizes non-punitive unjamming through local role recalibration, task restructuring, and communicative facilitation. When structural divergence proves persistent despite mediation, the framework acknowledges dignified ecosystemic relocation as a constructive option, safeguarding individual wellbeing and preventing chronic ostracism.

4. Tensorial Algebra, Q81 Lattice & Densities Pentagonram

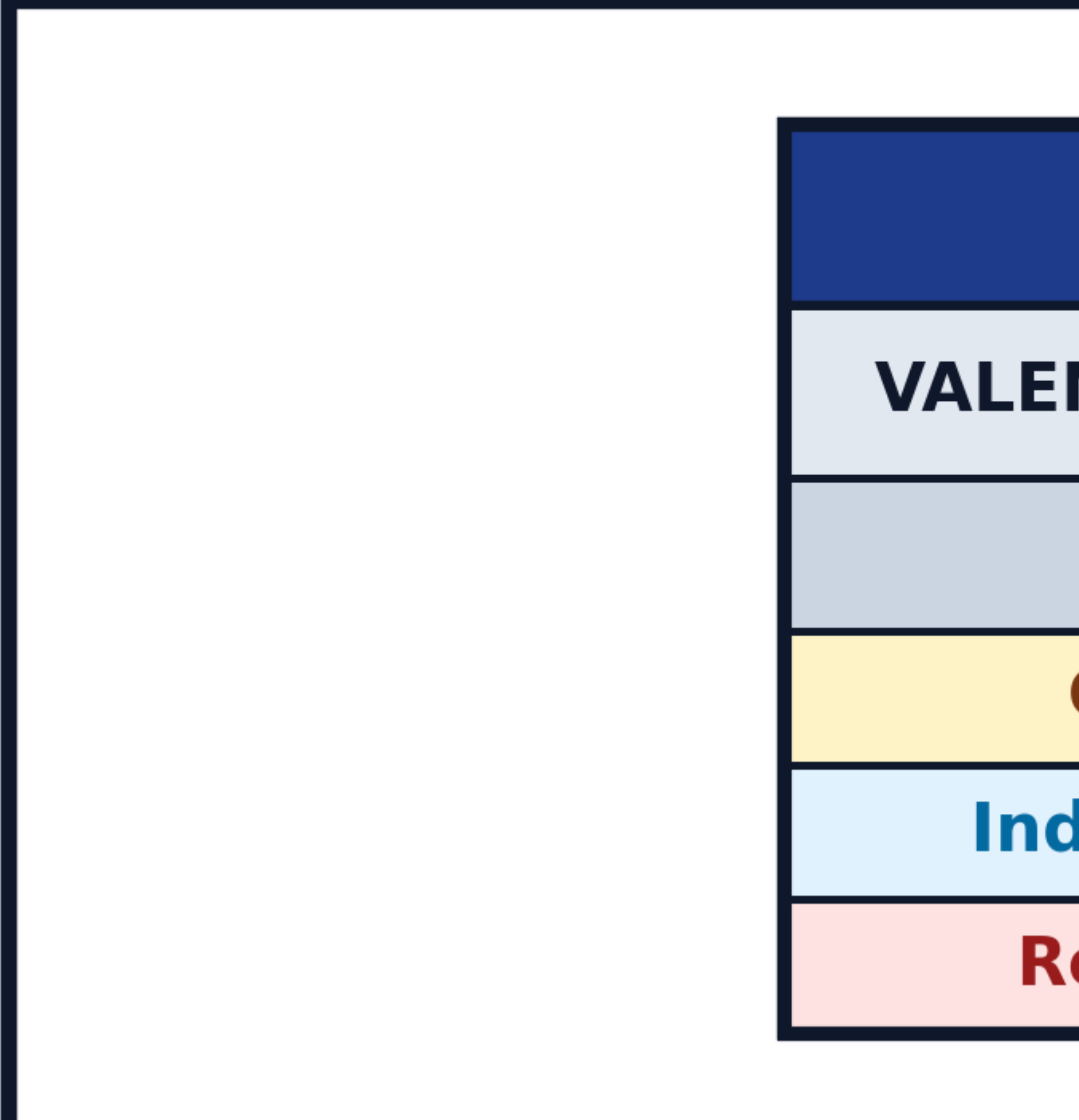
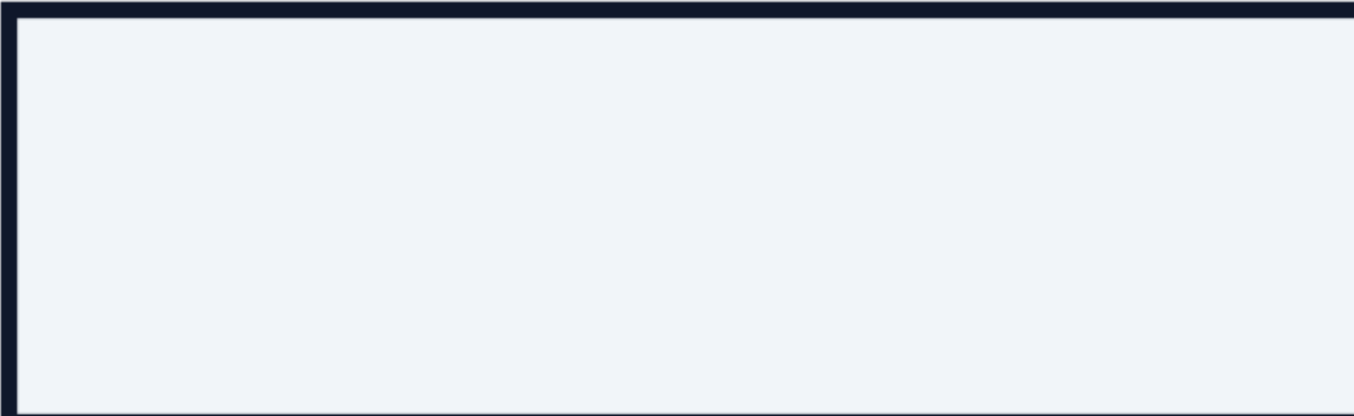
4.1. The Dyadic Tetragram & Dual Notation

Each cell (i, j) integrates four atomic moments: D1 = pDA (emitted expectation), D2 = DA (emitted choice), D3 = REC (received choice), and D4 = pREC (received expectation).

| Position / Dimension | Psychosocial Moment / Construct | Positive (+) [SMIb: A-Z] | Neutral (o) [SMIb: o]  | Negative (-) [SMIb: a-z] |
|----------------------|---------------------------------|--------------------------|------------------------|--------------------------|
| D_1                  | pDA (Emitted expectation)       | < (expects choice)       | i (expects neutrality) | [] (expects rejection)   |
| D_2                  | DA (Emitted choice)             | E (choice)               | ç (indifference)       | R (rejection)            |
| D_3                  | REC (Received choice)           | e (received choice)      | ? (received            | r (received              |

|            |                                         |                       |                              |                             |
|------------|-----------------------------------------|-----------------------|------------------------------|-----------------------------|
|            |                                         |                       | indifference)                | rejection)                  |
| <b>D_4</b> | <b>pREC</b> (Attributed<br>expectation) | > (attributes choice) | ! (attributes<br>neutrality) | ] (attributes<br>rejection) |

The lattice contains the 81 exhaustive relational configurations ( $Q_{81} = 3^4$ ), highlighting mutual choice ( $Q_{11}$ ), mutual rejection ( $[Rr]$ ), and reflexive indifference ( $[\ddot{r}r]$ ), decomposed analytically through 255 Boolean indicators ( $I_{255}$ ). This structured representation of signed relational patterns (+ / -) and dyadic balance connects fruitfully with contemporary research on signed graphs, communicability geometry, and structural frustration indices (Aref & Wilson, 2021; Cucuringu et al., 2023; Diaz-Diaz & Estrada, 2025).

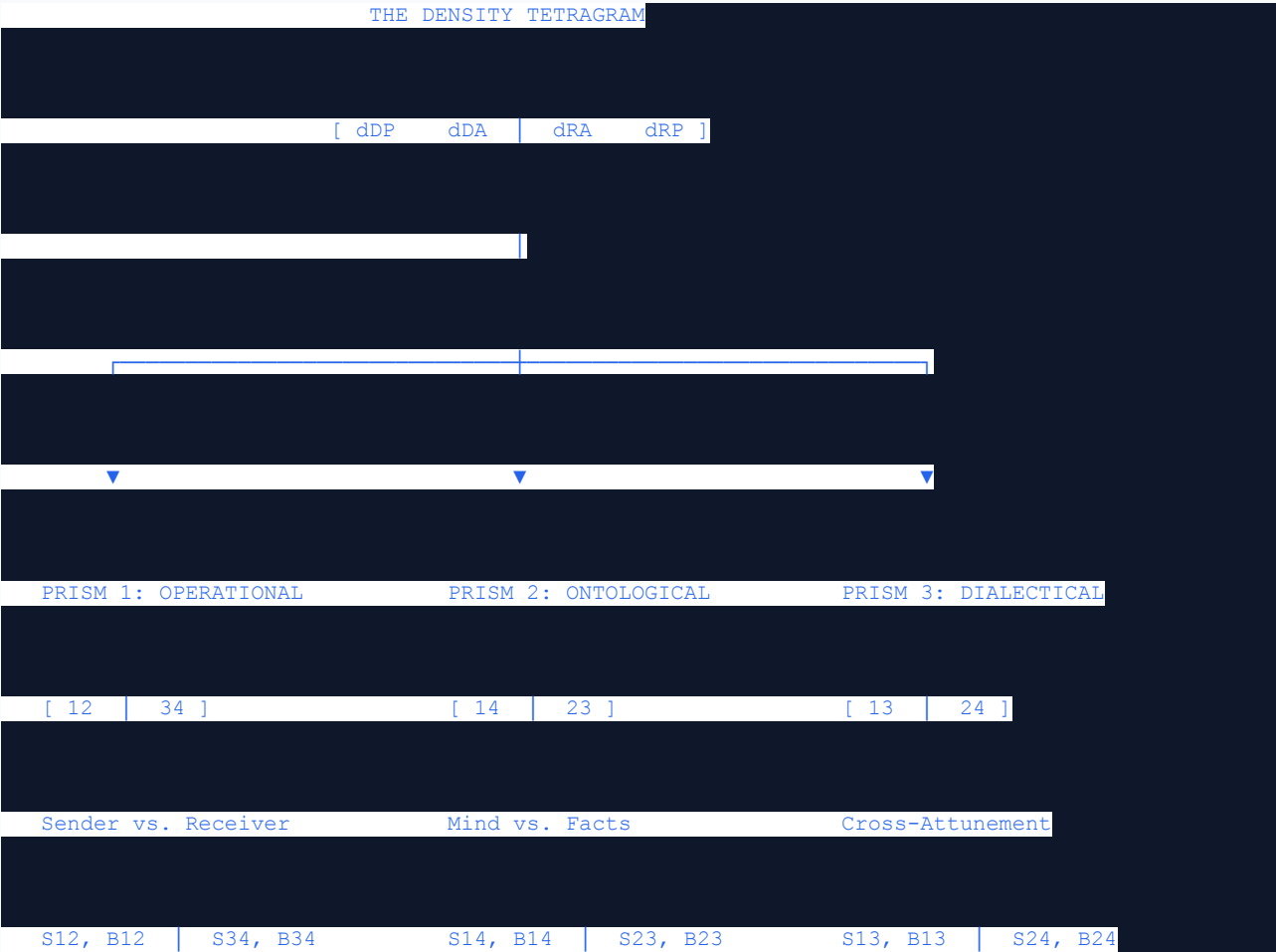


|             |
|-------------|
|             |
| <b>VALE</b> |
|             |
|             |
|             |
| <b>Ind</b>  |
| <b>Re</b>   |

4.2. The Canonical Doctrine of the 18 Relational Densities (18D) and the 3 Dual Prisms

Micro-interactional dyadic analysis transitions from discrete rank tallies to the socio-thermodynamic continuum through the **exponential relational decay law**  $W(k) = 2^{\{1-k\}}$  (\$1st=1.000, 2nd=0.500, 3rd=0.250...\$). To distinguish ordinal ranks from continuous physical magnitudes, the four components of the dyadic tetragram adopt the differential prefix **d**:  $\{dDP, dDA, dRA, dRP\} \equiv \{\mathbf{1}, \mathbf{2}, \mathbf{3}, \mathbf{4}\}$ . The continuous space operates under the universal prefix convention **\$S\$** (Signed Net Balance,  $\pm \mathbb{R}$ ), quantifying potentials and directional flux) and **\$B\$** (Unsigned Gross Mass,  $|\cdot| \geq 0$ ), quantifying total kinetic investment and metabolic energy).

The elemental quartet partitions into exactly **3 invariant dual prisms** ( $\binom{4}{2}/2 = 3$ ), providing exhaustive structural refractions of dyadic reality:



1. Prism 1: Operational Axis (Sender vs. Receiver)  $\rightarrow [12 \mid 34]$ :

Disentangles outbound expansiveness from inbound impact.

\* Sender Side (Outbound): Sender Balance  $S_{12} = dDP + dDA$  (SDA: net expansiveness) and Sender

Mass  $\$B_{12} = |dDP| + |dDA|$  (BDA: total emitted investment).

\* Receiver Side (Inbound): Receiver Balance  $\$S_{34} = dRA + dRP$  (SRC: induced status) and Receiver Mass  $\$B_{34} = |dRA| + |dRP|$  (BRC: convergent relational pressure).

Diagnostics:\* Uncovers power asymmetries, nurturing leaders, passive sinks, and structural ostracism.

### 1. Prism 2: Ontological Axis (Mind vs. Facts) $\rightarrow [14 \mid 23]$ :

Contrasts subjective expectations against objective interpersonal choices.

\* Mental World (Crossed Expectations): Perception Balance  $\$S_{14} = dDP + dRP$  and Perception Mass  $\$B_{14} = |dDP| + |dRP|$ .

\* Factual World (Effective Choices): Action Balance  $\$S_{23} = dDA + dRA$  and Action Mass  $\$B_{23} = |dDA| + |dRA|$ .

Diagnostics:\* If  $\$B_{14} \gg B_{23}$ , diagnoses **Collective Neurosis / Group Rumination** (energy dissipated on unfounded fears and cognitive fantasies); if  $\$B_{23} \gg B_{14}$ , diagnoses **Blind Impulsivity / Reflexive Anomie** (action devoid of empathy and mutual mentalization).

### 1. Prism 3: Dialectical Axis (Cross-Attunement & Realism) $\rightarrow [13 \mid 24]$ :

Crosses sender expectation with receiver action, auditing alignment and disappointment.

\* Sender Realism (Sender Expectation + Peer Action):  $\$S_{13} = dDP + dRA$  and  $\$B_{13} = |dDP| + |dRA|$ .

\* Receiver Response (Sender Action + Peer Expectation):  $\$S_{24} = dDA + dRP$  and  $\$B_{24} = |dDA| + |dRP|$ .

Diagnostics:\* Measures **latent relational friction and sociometric disillusionment** ( $\$S_{13} \ll 0$  when  $\$dDP > 0$ ).

Global system closure is established through Net Relational Potential ( $\$SDR = SDA + SRC$ ), Total Kinetic Mass ( $\$BDR = BDA + BRC$ ), and Positional Asymmetry ( $\$Dif-DR = SDA - SRC$ ), constituting the **18 continuous relational densities (18D space)**.

CONTINUO)

12

10

8



### 4.3. Continuous Quantification of the Individual Synthetic Relational Figure

Historically, aggregating an individual's collective standing through direct arithmetic averaging of discrete figures produced cancellation artifacts (blending high attractions and rejections yielded a false illusion of indifference). NEXORD resolves this by formalizing the **Individual Synthetic Relational Figure** via continuous vector decomposition:

- **Marginal Emission Density ( $\bar{d}_{i \cdot}$ ):** Mean outbound relational investment toward the network:  $\bar{d}_{i \cdot} = \frac{1}{N-1} \sum_{j \neq i} d_{ij}$  (mean expansiveness).
- **Marginal Reception Density ( $\bar{d}_{\cdot i}$ ):** Mean inbound social attraction converging upon the actor:  $\bar{d}_{\cdot i} = \frac{1}{N-1} \sum_{j \neq i} d_{ji}$  (mean status/in-centrality).
- **Integrated Global Density ( $D_i$ ):** Canonical semi-sum of marginal densities:

$$D_i = \frac{\bar{d}_{i \cdot} + \bar{d}_{\cdot i}}{2}$$

This index guarantees relational mass conservation ( $\frac{1}{N} \sum_i D_i = \Delta$ ), scaling the physical volume, radius, and envelope of the node glyph within 3D space.

- **Relational Differential Bias ( $\delta_i$ ):** Normalized giving/receiving asymmetry:

$$\delta_i = \frac{\bar{d}_{i \cdot} - \bar{d}_{\cdot i}}{\bar{d}_{i \cdot} + \bar{d}_{\cdot i}} \in [-1, +1]$$

where  $\delta_i > 0$  denotes expansive bias,  $\delta_i < 0$  denotes receptive bias, and  $\delta_i \approx 0$  marks symmetric reciprocity.

- **Flow Vector Angle ( $\theta_i$ ):** Directional angle on the (In, Out) plane:  $\theta_i = \arctan\left(\frac{\bar{d}_{\cdot i}}{\bar{d}_{i \cdot}}\right)$  ( $45^\circ$  = balanced reciprocity).

### 4.4. The VISORD Aquarium Paradigm & Relational Hydrostatics

The **VISORD Aquarium** paradigm translates the static 2D sociogram into a continuous 3D fluid volume. Actors float and interact according to socio-thermodynamic laws:

- **X Axis:** Emission / Expansiveness (\$SDA\$).
- **Z Axis:** Reception / Status (\$SRC\$).
- **Y Axis:** Hydrostatic Depth / Buoyancy ( $Z_i = D_i \times \text{scale}$ ).
- **Three Hydrostatic Strata:**

- 1. **Open Sky (Atmosphere):** Upper stratum for hyper-investing nodes exceeding water level ( $Z_i > Z_{\text{sea}}$ ).
  - 2. **Water Surface ( $Z_{\text{sea}}$ ):** Homeostatic group equilibrium plane.
  - 3. **Seabed / Benthos (Abyssal):** Basin of deficit buoyancy and severe ostracism ( $Z_i \rightarrow 0$  or  $BDF/BDC < 0.5$ ).
- **The Canonical VISORD Laboratories:** The platform deploys four specialized exploration suites: **VISORD\_demo** (phase transitions and 3D/4D dynamics), **VISORD\_cultural** (literary masterpieces with zero reactivity), **VISORD\_radiológico** (Grassmann topological screenings), and **VISORD\_dialéctico** (micro-interactions, social dilemmas, and IPA-SMIb Moviola), unified within the continuous volumetric hydrostatics of **VISORD\_acuario**.

## 5. Graphical Solutions, GxTyCz Multidimensional Design Space, and Invariant Geometry

### 5.1. Graphical Solutions and Parallel Dual Planes (AFC | PCA)

NEXORD deploys versatile graphical solutions to represent the continuous relational field with clarity and without information overload:

- **Three-Dimensional Continuous Sociograms:** Spatial rendering of attraction basins (neon green, ), opposition regions (orange), and friction zones (magenta,  $[Rr]$ ), equipped with longitudinal temporal controls.
- **Dual Factorial Space (AFC | PCA):**
  - \* **Correspondence Analysis (AFC):** Operates on relative contingency matrices via Chi-square ( $\chi^2$ ) metrics, ideal for categorical frequency profiles and sender/receiver polarization.
  - \* **Principal Component Analysis (PCA):** Operates on continuous density metrics and the inverse rank law ( $W = \pm 1/r$ ), preserving inertial mass and avoiding discrete ties.
  - \* **Commutable Dual Planes:** Disentangles observations into the Senders/Givers plane (who selects), the

Receivers plane (who is selected), or the unified **Factorial Biplot ( $F_1 \times F_2$ )**. Factor 1 (horizontal) summarizes Affective Polarization ( vs. [Rr]), whereas Factor 2 (vertical) projects Expansiveness and Power Visibility (Absolute Density BDR).

- **1D Slender Vertical Axes:** Vertical projections ranking nodes by structural weight, flanked by external diagnostic tables for absolute contribution (CTA), representation quality (COR), and cumulative variance (CTR).
- **Intergroup Structurograms:** Relational support polygons supporting comparative morphological diagnostics across teams.



10:14:13 AM 10/10/2023

SELECCIONAR EL NÚMERO DE PUNTOS

1000

Forma

Esfera

SELECCIONAR EL TIPO DE PUNTO

100

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SELECCIONAR EL TIPO DE PUNTO

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INICIAR DESDE CERO

EVOLUCIONAR A HORIZONTE

Presionar para salir

10:14:13 AM 10/10/2023

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VER

TABLA

SAIR DE PRESENTACIÓN

SELECCIONAR EL TIPO DE PUNTO

SELECCIONAR EL TIPO DE PUNTO

| 100 | 1000   | 10000  | 100000 | 1000000 |
|-----|--------|--------|--------|---------|
| 1   | -0.000 | -0.000 | 0.000  | 0.000   |
| 2   | -0.000 | -0.000 | 0.000  | 0.000   |
| 3   | -0.000 | -0.000 | 0.000  | 0.000   |
| 4   | -0.000 | -0.000 | 0.000  | 0.000   |

ACTIVADO AUTOMÁTICO

100

3D CARTESIAN SYSTEM

X:  $\pm SDR$   $[-1.0 \leftrightarrow +1.0]$

Y: BDR  $[0.0 \rightarrow 150.0]$

Z:  $\Phi$   $[0.0 \rightarrow 100.0]$

$\leftarrow -SDR$  (Rejection)

-1.0

$\Phi 100$   
Z AXIS: RELAT

Figure 4: MESO Factorial Biplot ( $F_1 \times F_2$ ) in Parallel Dual Space (Benzécri AFC // 10D Socio-Thermodynamic PCA). Left Panel: AFC on discrete contingency  $\mathbb{N}_0$  and Q81 invariant lattice. Right Panel: Continuous 10D PCA ( $\pm\mathbb{R}$ ) with the Densities Pentagon, positive vectors in  $+X$  and dashed projective ray extensions to  $-X$  passing through the origin  $(0,0)$ . Both spaces project actors as dyadic vectorial dipoles  $[DA \longrightarrow RC]$  with central barycenters (+), revealing the attraction basin (2 & 1), the homeostatic bisectrix (3 & 4), and the ostracism/entropic dissipation basin (6 & 5).

## Plano F

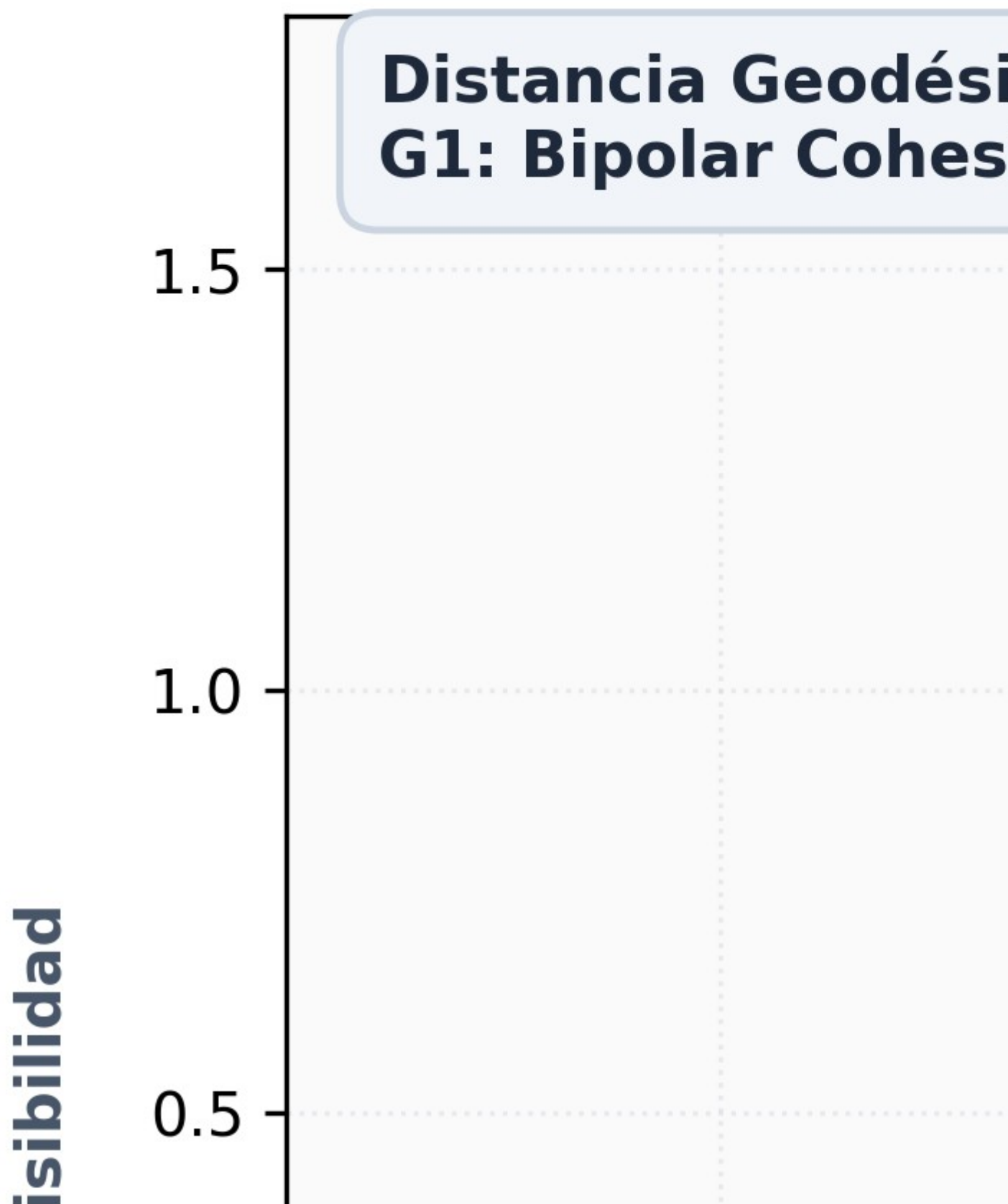


Figure 5: Intergroup Comparative Factorial Plane (AFC / PCA). Polarization force vectors and polygonal relational support structurograms between two independent collectives.

### 5.1.1. Paradigm Shift: From Categorical Contingency (Moreno / Benzécri) to the Socio-Thermodynamic Continuum (10D PCA)

The parallel articulation of Correspondence Analysis (AFC) and 10D Socio-Thermodynamic Principal Component Analysis (10D PCA) analytically formalizes the profound paradigm shift that Computational Ordinal Sociometry introduces beyond Moreno's classical sociometry (1934) and Benzécri's contingency approach (1973):

- **1. Ontological Metric Nature: Categorical Discretization ( $\mathbb{N}_0$ ) vs. Continuous Energy Continuum ( $\mathbb{R}$ ):**

Classical Sociometry (Moreno, 1934; Benzécri, 1973):\* Operates on contingency tables constructed from discrete frequency tallies ( $\mathbb{N}_0$ ) constrained by fixed quotas (typically 3 choices and 3 rejections). The Chi-square metric ( $\chi^2$ ) divides observed counts by the square root of marginal mass products ( $\sqrt{r_i \cdot c_j}$ ), artificially inflating peripheral low frequencies and dispersing factor inertia across multiple orthogonal dimensions ( $F_1 \approx 50.7\%$ ,  $F_2 \approx 37.1\%$ ). Most critically, AFC reifies mutual rejection  $[Rr]$  as a substantive, autonomous qualitative pole symmetric to choice.

NEXORD Socio-Thermodynamics (2026):\* The continuous exponential rank decay operator  $W(k) = 2^{1-k}$  ( $1\text{st}=1.000$ ,  $2\text{nd}=0.500$ ,  $3\text{rd}=0.250\dots$ ) applied over tie-free forced drag-and-drop administration transforms ordinal rankings into continuous signed densities ( $\mathbb{R}$ ). Within 10D PCA space (orthogonally decomposed via SVD on the centered Pentagon matrix), all 10 canonical density variables (SDR, BDR, SDA, SRC, BDA, BRC, D<sub>1</sub>..D<sub>4</sub>) converge into the positive half-plane ( $+X$ ). Mutual rejection  $[Rr]$ , conflict, and exclusion are no longer an autonomous qualitative pole, but an **energy deficit, gravitational void, and loss of relational mass**, mapped onto the negative projective ray extensions ( $-X$ ) passing through the barycentric origin  $(0,0)$ .

- **2. From Static Scalar Node to Dynamic Dyadic Dipole ( $DA \leftrightarrow RC$ ):**

\* In classical sociograms, the individual is depicted as a static scalar dot devoid of mass or physical work, or as a passive centroid in factorial space.

\* In NEXORD, the relational actor is ontologically modeled as a **kinematic vectorial dipole in continuous tension**: their active outbound investment ( $DA / SDA$ ) and inbound collective reception

(\$RC / SRC\$) are projected simultaneously. The directed vector arrow \$DA \rarr\$; RC\$ quantifies socio-thermodynamic work and positional asymmetry:

$$\Delta_i = RC_i - DA_i$$

\* The actor's intrinsic barycenter (\$+\$) sits with mathematical exactitude at the vector midpoint \$[(DA + RC)/2]\$, preserving individual systemic identity without static collapse.

- **3. The Three Topological Field Basins:**

Gravitational Attraction Basin (Leaders 2 & 1):\* Upward vectors with strongly positive balances (\$\Delta > 0\$). The group invests massive gravitational mass in them without requiring sender hyper-effort (\$RC \gg DA\$). They function as gravitational potential wells that structure, polarize, and hold the collective network together at zero entropic cost.

Homeostatic Bisectrix & Critical Freezing Threshold (Intermediate Actors 3 & 4):\* Subject 3 sits precisely on the homeostatic bisectrix (\$RC = DA\$, \$\Delta \approx 0\$), acting as a thermal damper and balancing mediator that buffers systemic fluctuations. Subject 4 marks the critical threshold of relational freezing: receiving zero positive impact (\$RC = 0\$), yet sustaining the group through emitted effort (\$DA > 0\$), accumulating severe strain (\$\Delta < 0\$).

Ostracism & Entropic Dissipation Basin (Peripheral Actors 6 & 5):\* Collapsed, downward-plunging vectors with heavily negative balances (\$\Delta < 0\$). Positioned entirely within the negative projective half-plane (\$-X\$), diametrically opposite to all density vectors (\$SDR, SRC, BDR...\$). They act as entropic dissipation sinks absorbing collective frictions and tensions, exposing structural exclusion invisible to classical sociometric tallies.

- **4. Inertial Dominance of the Gravitational Gradient (\$PC\_1 \gg PC\_2\$):**

\* In sharp contrast to AFC inertia dispersion, in socio-thermodynamic PCA the first principal component (\$PC\_1\$) accounts for over **\$87.5\%** of total inertia (and up to \$90\%\$ across longitudinal field corpora).

\* This analytical result demonstrates that human collectives follow a **unified law of socio-affective gravity**: the relational field is not fragmented into disjointed axes, but organized along a primary dipolar gradient separating concentrated relational density (\$+X\$) from dissipative relational voids (\$-X\$).

- **5. The Barycentric Conservation Theorem:**

\* In any closed human collective, the fundamental conservation law holds analytically and empirically:

$$\sum_{i=1}^N DA_i \equiv \sum_{j=1}^N RC_j$$

\* Consequently, the collective mass centroid converges with floating-point precision ( $\Delta_{\text{centroid}} = 0.000$ , machine epsilon  $\approx 2.22 \times 10^{-16}$ ) exactly at the Cartesian origin  $(0,0)$ . All relational energy received by an actor ( $RC$ ) is strictly conserved from outbound peer investments ( $DA$ ): no relational mass is created or destroyed; it is merely redistributed across the dyadic field tensor.

## 5.2. The GxTyCz Multidimensional Design Space without a Priori Constraints

The mathematical formulation of NEXORD imposes no a priori limitations or restrictions on research and diagnostic designs:

- **Groups (G):** Accommodates any number of independent collectives, organizational departments, working teams, or experimental groups.
- **Temporal Waves (T):** Enables continuous longitudinal tracking across T timepoints ( $T_1, T_2, \dots, T_n$ ), monitoring the diachronic evolution of cohesion and latent structural rifts.
- **Multiplex Criteria (C):** Models multiple functional and affective criteria simultaneously ( $C_1$  task competence,  $C_2$  social bonding,  $C_3$  crisis leadership), capturing topological shifts across evaluative lenses.
- **Contextual Variables (Z):** Seamlessly integrates demographic profile attributes (age, gender, tenure) and group activity scales (AAG; Vicente, Cornejo & Barbero, 2006).

Alongside core ordinal rankings, the system's formal architecture articulates two complementary psychometric modules:

- **Native Triadic Metaperceptions ( $aE$  and  $oE$ ):** To complete the actor's socio-cognitive field without inflating administrative burden, NEXORD natively captures triangulated expectations projected onto the boundary poles of the relational constellation:

\*  **$aE$  (Expectations on the Closest Peer):** Who does the sender expect their closest peer (1st choice) will prefer to choose?

\*  **$oE$  (Expectations on the Most Distant Peer):** Who does the sender expect their most distant or rejected peer will prefer to choose?

This triadic attribution operationalizes socio-cognitive closure (following Heider and Festinger), assessing perceived transitivity, cognitive balance, and structural coherence within the group.

- **Modular Reference Batteries ( $V_{10}$  and  $A_{16}$ ):** To ensure cross-group metric standardization within the Sociometric Data Bank (BSOC), the platform specifies by default a decadic sociodemographic battery ( $V_{10,3}$ , ten key attributes with three modalities) and the Group Adjective Attribution inventory ( $A_{16,4}$ , 16 canonical adjectives structured across four factors: Thematic, Functional, Cognitive, and Affective, utilizing 1-8 forced even scales). Nonetheless, both batteries remain modular, optional reference baselines: they can be modified, replaced, or tailored under the Principal Investigator's (PI) criteria according to specific substantive requirements.

### 5.3. Multigroup PCA & Geodesic Distance on Grassmann Manifolds $Gr(k, N)$

To compare formally independent collectives without enforcing artificial cross-group links, each group is modeled as a  $k$ -dimensional subspace in  $R^N$ . From canonical Jordan angles  $\cos(\theta_m) = \sigma_m(U_1^T U_2)$ , scale-invariant geodesic distance is calculated:

## Formula 7: Geodesic Distance on Grassmann Manifolds

This metric quantitatively determines whether two collectives share structural morphology regardless of sample size differences.

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## 6. Experimental Research: Physical Lab, Dialectical Sociometry, and Multimodal Diagnostics

### 6.1. Monitoring Experimental Designs in the Physical Laboratory

The platform supports the systematic monitoring of face-to-face experimental designs within physical social psychology laboratories. In controlled observation rooms, working seminars, and team behavioral laboratories, NEXORD captures interpersonal dynamics in real time, tracking the emergence of informal leadership structures, reciprocity, and network centrality during collaborative tasks.

### 6.2. Dialectical Sociometry and Social Dilemmas

Conceived as an **emerging experimental research track currently in active development**, Dialectical Sociometry investigates the real-time recording of continuous interaction sequences. Through the controlled delimitation of an **interaction space** and calibrated **interaction counters** (speaking turn frequencies, conversational pacing, and mutual listening balances), this modeling framework observes interactional dynamics in social dilemmas and collaborative tasks, monitoring the emergence of trust, gradual reciprocity, or free-riding under strict anonymization protocols.

### 6.3. Multimodal Perception: The Iceberg Model & Hydrodynamic Sonification

In all human groups, a visible surface layer (formal roles, explicit agreements, observable behavior) is sustained by a deep, submerged relational foundation (unspoken expectations, latent tensions, unvoiced affinities, and subtle rejections). The fundamental methodological objective of the platform is to **render visible and audible the dynamics and processes of group interaction**:

- **Visual Dimension:** Articulated through continuous graphical solutions and 3D sociograms.
- **Auditory Dimension (Hydrodynamic Sonification):** Translates friction gradients ( $\Phi$ ), net

potential (SDR), and flow asymmetries into harmonic acoustic frequencies (calibrated at 432 Hz with distinct timbral textures). This auditory mapping allows researchers to listen to relational tension or harmony, providing a complementary perceptual modality for detecting dynamics submerged beneath the visible surface.

PERCENT ACTIVITY

68

PLI20

44

EXP. PROB. (%)

0.61 →

19

### 3D FIELD VOLUME

X:  $\pm$ SDR [-1.0  $\leftrightarrow$  +1.0]

Y: BDR [0.0  $\rightarrow$  250.0]

Z:  $\Phi$  [0.0  $\rightarrow$  200.0]

← -SDR (Rejection)

-1.0

$\Phi$

## 6.4. The CULTURAL Corpus as a Zero-Reactivity Laboratory

Through the CULTURAL data category, NEXORD transcribes dramatic, cinematic, and narrative masterpieces into ordinal sociomatrices. Serving as complete and finalized relational records, literary corpora enable the forensic analysis of extreme dynamics (such as polarization or authoritarianism in works by Lorca, Sartre, or Rose) within an experimental framework with zero evaluation reactivity.

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## | 7. Institutional Governance, Ethics by Design, and Open Science

### 7.1. Ethics by Design & Non-Punitive Interdiction

The platform embeds ethical safeguards directly into technical algorithms (*Ethics by Design*). The system explicitly prohibits punitive deployments, negative selection, or elimination rankings. Analysis on empirical field datasets (REAL category) requires authentication and formal accountability by an accredited Principal Investigator (PI).

- **Explicit Deontological Mandate of the Principal Investigator (PI):** It is strictly prohibited to attribute strictly group socio-thermodynamic phenomena and field positions to intrinsic individual personality traits or personal merit/demerit. All sociometric diagnostics and interventions must safeguard the personal self-esteem and social reputation of participants, steering intervention toward environmental recalibration, role clarification, and collective wellbeing.

### 7.2. Open Science, Clélie Protocol & Institutional Custodianship (BSOC)

In accordance with Open Science and FAIR Data principles, NEXORD's mathematical, methodological, and computational formulations are publicly accessible. The complete interactive web platform, computational engines, and 3D VISORD visualizers are openly accessible at the official web portal: <https://josemancor.github.io/nex-ord-web/>. To protect participant confidentiality, empirical field datasets undergo canonical anonymization (1A1a..5A1a) and cryptographic hash sealing (SHA3-512, Clélie Protocol), under the academic custodianship of the Sociometric Data Bank (BSOC / University of Barcelona).

### 7.3. Participant Rights & Constructive Feedback

Participants receive individual codes to audit data processing fidelity without accessing peer records. Node re-identification is strictly restricted to the responsible PI within guidance or supportive group mediation. All feedback is constructive, non-stigmatizing, and dedicated to promoting collective health.

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## 8. Canonical Technical Annexes

### 8.1. Annex I: Dual Matrices (SMIb/SMIa), Master 9x9 Lattice of Q81, and I255 Catalog

- **SMIb Matrix:** Stores the numerical tetragram  $(A_1 \cdot A_2 \cdot A_3 \cdot A_4) = (pDA \cdot DA \cdot RECIBE \cdot pRECIBE)$  in each cell  $(i, j)$ .
- **SMIa Matrix:** Condenses the tetragram into the 81 dyadic configurations  $(1 = [Rr], 41 = i\dot{c}?! , 81 = )$ .
- **Boolean Indicator Catalog I255:** Disjunctive partitioning of dyadic space along the diagonal of perceptual consonance.

### 8.2. Annex II: Factorial Quality Metrics (COR, CTA, CTR), Vertical Slender Axes, and 2D(6) Projection

- **Factorial Metrics (Cornejo, 1988):**

\* COR measures representation quality of node  $i$  on factor  $\alpha$ .

\* CTA expresses the percentage of factor inertia explained by element  $i$ .

\* CTR reflects global cumulative explained variance.

- **Vertical Factorial Axes:** Slender vertical axes rank nodes by structural weight, preventing visual crowding in saturated 2D scatter plots.

## Eje Factorial 1D Equilibrio B



1A (Subg



2A (Subg



3A (Subg

### 8.3. Annex III: Catalog of the 10 Ecosystemic Domains

NEXORD calibrates Relational Inertia ( $\alpha$ ) and Relational Decay ( $\gamma$ ) across 10 normative institutional domains:

# Table Annex III: Cal

*Thermodynamic Calibration Para*

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No.

#### 8.4. Annex IV: Sender - Receiver Rank Interdistances and Isometry Diagonal

Euclidean distances for sender emission ( $d_{DA}$ ) and receiver reception ( $d_{RC}$ ) preserving strict sender ordering:

- **Isometry Diagonal ( $d_{DA} = d_{RC}$ ):** Evaluates relational symmetry and verifies the Relational Asymmetry Principle (reception dispersion systematically exceeds emission dispersion).

## Matriz de

**1A** - **0.00**

**2A** - **0.85**

**3A** - **1.16**

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## 9. Canonical References

1. Absil, P.-A., Mahony, R., & Sepulchre, R. (2008). *Optimization algorithms on matrix manifolds*. Princeton University Press.
2. Aref, S., & Wilson, M. C. (2021). The frustration index is a key measure for analysing signed networks. *Journal of Complex Networks*, 7(2), cnz006. <https://doi.org/10.1093/comnet/cnz006>
3. Avramidis, E., Strogilos, V., Aroni, K., & Kankaraki, C. T. (2017). Using sociometric techniques to assess the social impacts of inclusion. *Educational Research Review*, 20, 68–80.
4. Bales, R. F. (1950). *Interaction process analysis: A method for the study of small groups*. Addison-Wesley.
5. Benzécri, J.-P. (1973). *L'Analyse des Données: T. 2, L'Analyse des Correspondances*. Dunod.
6. Borsboom, D., et al. (2021). Network analysis of multivariate data in psychological science. *Nature Reviews Methods Primers*, 1(1), 58.
7. Cornejo, J. M. (1988). *Técnicas de investigación social: El análisis de correspondencias*. PPU.
8. Cornejo, J. M. (2026). *Computational Ordinal Sociometry (NEXORD): An Algorithmic, Matrix, and 4D Socio-Thermodynamic Framework*. Zenodo. <https://doi.org/10.5281/zenodo.18926396>
9. Cucuringu, M., Mantzaris, A. V., & Roberts, J. (2023). The Bradley-Terry stochastic block model for directed ranking networks. *Journal of Complex Networks*, 11(4), cnad025. <https://doi.org/10.1093/comnet/cnad025>
10. Diaz-Diaz, F., & Estrada, E. (2025). Signed graphs in data sciences via communicability geometry. *Information Sciences*, 680, 121110. <https://doi.org/10.1016/j.ins.2024.121110>
11. Festinger, L. (1950). Informal social communication. *Psychological Review*, 57(5), 271–282.
12. Heider, F. (1958). *The psychology of interpersonal relations*. John Wiley & Sons.
13. Janis, I. L. (1972). *Victims of Groupthink: A psychological study of foreign-policy decisions and fiascos*. Houghton Mifflin.
14. Kenny, D. A., & Garcia, R. L. (2012). Using the Actor–Partner Interdependence Model to study the effects of group composition. *Small Group Research*, 43(4), 468–496.
15. Krivitsky, P. N., & Butts, C. T. (2017). Exponential-family random graph models for rank-order relational data. *Sociological Methodology*, 47(1), 68–112. <https://doi.org/10.1111/socm.12038>
16. Lewin, K. (1951). *Field theory in social science: Selected theoretical papers*. Harper & Row.
17. Morales Domínguez, J. F., & Cornejo Álvarez, J. M. (2026). Sociometría Ordinal Computacional. La Plataforma NEXORD: Hacia una Física de lo Grupal. *Psicothema* (in review).

18. Moreno, J. L. (1934). *Who shall survive? A new approach to the problem of human interrelations*. Nervous and Mental Disease Publishing Co.
19. Nicolis, G., & Prigogine, I. (1977). *Self-organization in nonequilibrium systems: From dissipative structures to order through fluctuations*. John Wiley & Sons.
20. Scudéry, M. de (1654). *Clélie, histoire romaine*. Augustin Courbé.
21. Terry, R. (2000). Recent advances in measurement theory and the use of sociometric techniques. In A. H. N. Cillessen & W. M. Bukowski (Eds.), *Recent advances in the measurement of acceptance and rejection in the peer system* (pp. 27–48). Jossey-Bass. <https://doi.org/10.1002/cd.23220008804>
22. Tsankova, E., & Tair, E. (2021). Meta-accuracy in sociometric status perceptions. *Frontiers in Psychology*, 12, 642398.
23. Vicente, R., Cornejo, J. M., & Barbero, F. (2006). Análisis de la Actividad Grupal (AAG). *Psicothema*, 18(3), 445–452.
24. Ye, K., & Lim, L.-H. (2016). Schubert varieties and distances on Grassmann manifolds. *SIAM Journal on Matrix Analysis and Applications*, 37(3), 1176–1197.

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